Lab : 13

understanding the Architecture of pre-trained model.

Aim: To understand the Arch of pre-trained model

Objective:

* To learn the concept of transfer learning and how it reduces training time.

* To explore the layer wise architecture of pre trained models such as VGG16, AlexNet.

* To identify the role.

* To visualize how these models extract features

Pseudocode:

1, import libraries

2, Load a pre-trained model

3, model display it.

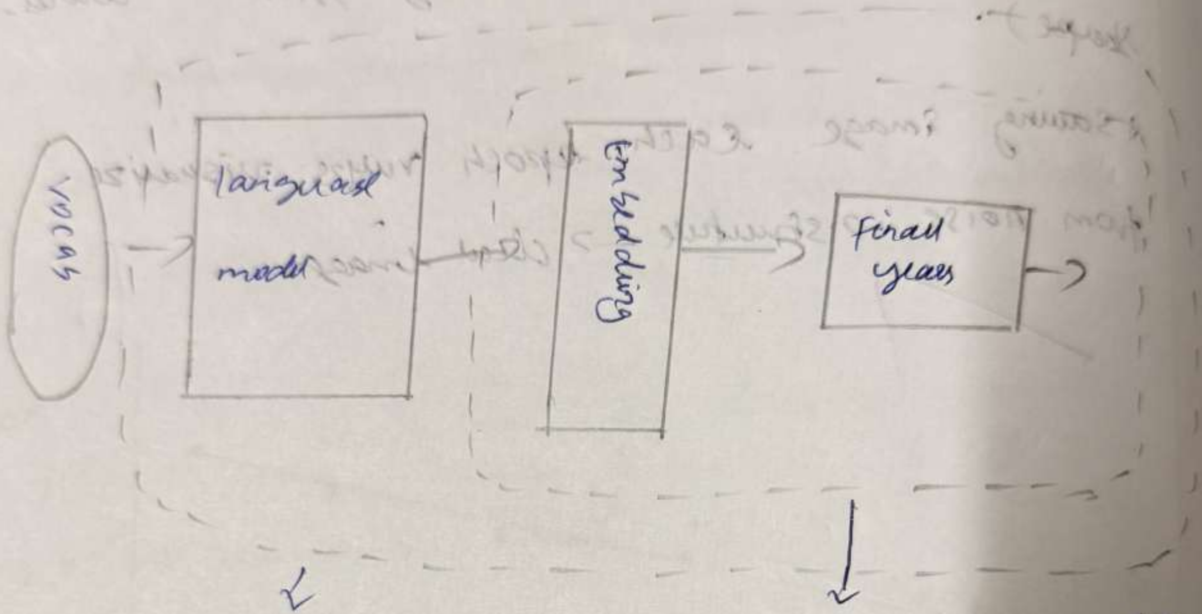
4, Observe: Input and output shapes layers name Types and no. of parameters.

5, Visualize the Architecture using:

6, Load a sample input image and preproces it for model

7, Pass image through the model and get predi

8, optionally visualize activation



language model
fine - tuning

pre-trained embedding
model.

Observation:

+ The pre-trained model contain millions of parameter spread across convolution & dense layer

+ Early layer extract edges and texture while deeper layers capture high level features such as object parts.

+ The use of pre-trained weights drastically reduces training time and improves model accuracy for new tasks.

Results:

Successfully Implemented pre-trained mod output

Linear (in-features = 2048, out features = 2 bias = True

VGG (

(features) : sequential (

(0) Conv 2d (3, 64, kernel size) = (3, 3), stride = (

padding = (1, 1))

1, Relu (inplace = True)

2, conv2d (64, 64 , kernel size = (3, 3), stride = (1, 1

padding (1, 1).

3, Relu (inplace = True)

4, maxpool 2d (kernel size) = 2 , stride = 2, pad

dilation = 1 ceil = mode = False) :

Total trainable parameters : 13835 7542